Finite propagation: scalar rigidity, internal states and record limits

Two scalar rigidity statements hold with exact ballistic support. A probability convolution semigroup on the line supported in $[-ut, ut]$ is deterministic drift. A strongly continuous translation-invariant unitary group on $L^2(\mathbb{R})$ with kernels supported in $[-u|t|, u|t|]$ is translation at some $|a| \le u$, up to a constant phase. The proofs below use variance additivity and the imported Paley–Wiener–Schwartz theorem, respectively. These are internal written derivations, not independent peer review or a claim of literature novelty.

Nontrivial bounded-speed dispersion can be carried by internal states. The two-state identities, hidden slow-mode counterexample and specified readout-noise limit below do not classify all matrix semigroups or force an action scale. The historical cone/arrow comparison is maintained in source evidence and interfaces.

Scalar probability kernels

C032. For probability measures, if $\mu_0 = \delta_0$, $\mu_{s+t} = \mu_s * \mu_t$ and $\operatorname{supp}\mu_t \subset [-ut, ut]$, then $\operatorname{Var}\mu_T = n \operatorname{Var}\mu_{T/n} \le u^2T^2/n \to 0$ and $\mu_t = \delta_{bt}$ with $|b| \le u$. Adjacent sections of such a law are translates of one another. The telegraph process escapes the theorem because its position alone is not Markov: its kernel's Fourier transform $e^{-\lambda t}[\cosh(st) + (\lambda/s)\sinh(st)]$ with $s = \sqrt{\lambda^2 - u^2k^2}$ is an even function of s, hence entire in k, but it is not a convolution semigroup in t.

Scalar unitary kernels

Proposition. Let $(U_t)_{t\in\mathbb{R}}$ be a strongly continuous one-parameter unitary group on $L^2(\mathbb{R})$ that commutes with translations, and suppose that for every t the kernel K_t with $U_t\psi = K_t * \psi$ has support in $[-u|t|, u|t|]$. Then there are real constants a, b with $|a| \le u$ such that $U_t\psi(x) = e^{-ibt}\psi(x - at)$.

Proof. Commutation with translations makes U_t a Fourier multiplier by a unimodular measurable $m_t(k)$, and Stone's theorem gives $m_t(k) = e^{-it\omega(k)}$ for a real measurable ω. The support hypothesis and the Paley–Wiener–Schwartz theorem extend each m_t to an entire function with $|m_t(k)| \le C_t(1 + |k|)^N e^{u|t||\operatorname{Im} k|}$. The identity $m_t m_{-t} = 1$ holds on the real axis, hence on $\mathbb{C}$, so m_t has no zeros and $G_t = \log m_t$ is entire. For fixed t, the bounds for m_t and m_{-t} give $|\operatorname{Re} G_t(z)| \le u|t||\operatorname{Im} z| + N_t \log(1 + |z|) + C_t$. For its harmonic real part, second-derivative estimates on a disk of radius R are $O((R + \log R)/R^2)$ and vanish as $R \to \infty$. Thus the real part, and hence G_t, are affine; its value on the real axis is zero. Thus $m_t(k) = e^{-i(a_t k + b_t)}$, with real a_t, b_t and $|a_t| \le u|t|$. The group law makes a_t additive; the bound implies $a_t = at$. Strong continuity makes the remaining phase a continuous character, hence e^{-ibt}. This proves the form without selecting logarithms continuously in time. □

Reading. Scalar evolutions in this class are translations at any $|a| \le u$, with an arbitrary constant phase rate. The massless chiral choices $a = \pm u$ saturate the speed bound. Interpreting phase rate as energy requires an action conversion factor. The Dirac evolution keeps light-cone support and unitarity by carrying two components; its scalar reduction has $\omega(k) = \sqrt{u^2k^2 + \omega_0^2}$, which is real on the axis but has branch points at $k = \pm i\omega_0/u$ and so cannot be the dispersion of a scalar kernel of compact support. At $u = c$ and $\omega_0 = mc^2/\hbar$ the branch points sit at the inverse reduced Compton length. Matching a telegraph rate to ω_0 makes their reciprocal scales coincide algebraically. This is a chosen model identification; analytic continuation alone identifies neither probabilities nor physical dynamics.

Bounded-speed fluctuations and the double limit

Refining time cuts and position resolution are distinct limits. For the symmetric telegraph model with mass m, speed u and switching rate $\lambda > 0$, put $H_* = mu^2/\lambda$ and $\Delta_* = 1/\lambda$. The window observable obeys $\mathsf{h}(\Delta) \le H_*\Delta/\Delta_*$ (finite-speed response), and the conditioned midpoint obeys the exact law $\kappa_{\mathrm{mid}}/H_* = g(\lambda T) = zR(z)[1 - R(z)]$ of the return-bridge calculation, with cubic onset. The fixed-preparation short-window intrinsic coefficient vanishes under a speed ceiling; the specified crossover window $\Delta_* = H_*/(mu^2) = 1/\lambda$ concerns a long-window plateau, and g describes its approach. This does not assert a limit for noisy records whose error scale changes with the cut. Which premise makes the plateau positive and common to all bodies is A08's question, not a consequence of scalar finite propagation.

Internal states, a matrix conjecture and readout noise

The scalar proposition is **not** a classification of matrix-valued finite-speed semigroups. A candidate, still unproved, says that a translation-invariant, strongly continuous contraction semigroup on $L^2(\mathbb{R};\mathbb{C}^n)$ whose convolution kernels have support in $[-ut, ut]$ has Fourier multipliers $e^{t(B-ikA)}$, where $A = A^\dagger$, $\|A\| \le u$ and $B + B^\dagger \le 0$. The missing steps include pointwise time continuity from strong operator continuity, a justified matrix logarithm and differentiation of its generator. Positivity alone does not make B a conservative Markov switching generator: $T_t f = e^{-t} f$ is positive and has zero propagation but loses mass. Conservation must be required separately. The scalar proof above does not discharge these obligations.

For the symmetric two-state telegraph generator $Q = \lambda(\sigma_x - I)$ and $V = u\sigma_z$, the exact matrix identity is $\|[V, Q]\| = 2u\lambda$, while the stationary velocity plateau is $H_* = mu^2/\lambda = 2mu^3/\|[V, Q]\|$. A coherent two-component model with $A = c\sigma_z$, $B_H = \omega_0\sigma_x$ similarly has $\|[A, B_H]\| = 2c\omega_0$. Converting frequency into rest energy uses a **supplied** action K, $m = K\omega_0/c^2$; physical $\hbar$ enters only if separately identified with K. A commutator norm does not see every slow mode: the four-state counterexample keeps the plateau fixed as an independent hidden label's relaxation gap closes. For $Q = 0$, a random constant velocity $\pm u$ instead has $\mathrm{Var}(X_\Delta - X_0) = u^2\Delta^2$; a deterministic velocity has zero variance. Commuting velocity sectors therefore do not justify the old general zero-response claim.

There is also an **exact, model-dependent readout effect**. Define $\mathsf{h}(\Delta) = m\,\mathrm{Var}(X_\Delta - X_0)/\Delta$ for a fixed bounded-speed motion. At the two endpoints add independent centered errors, independent of the motion, each of variance $\varepsilon^2/12$. The observed increment then satisfies

$$\mathsf{h}_\varepsilon(\Delta) = \mathsf{h}(\Delta) + \frac{m\varepsilon^2}{6\Delta}.$$

For fixed $\varepsilon > 0$, the $\Delta \downarrow 0$ limit is infinite; for fixed $\Delta > 0$, removing the readout error first and then refining gives zero by finite speed. Along $\varepsilon^2 = 6\kappa_0\Delta/m$, the observed coefficient tends to any chosen $\kappa_0 \ge 0$. Independent dither or additive-noise measurement is a premise here; fixed-lattice deterministic rounding need not satisfy it. On a uniform N-cell partition, pin endpoint errors to zero and give the $N - 1$ interior errors this variance. Expanding the kinetic-action squares, the expected added action is $m(N-1)\varepsilon^2/(12\Delta)$, with correlated neighboring increment errors accounted for in the expectation. This scaling selects a **readout bias**, not a lower bound on intrinsic action and not physical $\hbar$.

Exact free-kernel composition

The independently useful time-refinement manuscript and PDF retain the normalized free-kernel proof, Rivero's 1998 intermediate-coordinate proposal and the separate Brouder/Cariñena source leads. Their hypotheses are not those of the ballistic-support theorems above.

In that proof a positive Gaussian kernel of variance $\kappa t/m$ composes exactly under integration over an inserted position for every $\kappa > 0$. Weak continuity and Gaussian time homogeneity

permit every $\kappa = ma \geq 0$; $a = 0$ is the deterministic Dirac member. Requiring a positive density rules out zero by premise. Passing from positive kernels to oscillatory unitary propagation is another chosen step, and neither step identifies κ with physical $\hbar$. The Newton synthesis distinguishes this free fixed-line result from the two-regulator RG question.

Consequence for STATE

Scalar finite-speed rigidity, the open matrix classification and the specified readout double limit remain separate. At fixed preparation, bounded speed makes the intrinsic short-window coefficient vanish; independent readout errors can create any prescribed resolution-dependent coefficient. Exact Gaussian composition also permits a deterministic zero member. None of these statements supplies a universal positive physical action, quantum reconstruction or historical transmission claim.